

**MINISTRY OF EDUCATION AND TRAINING
FPT UNIVERSITY**

Default Risk Prediction in Banking

by

Nguyen Duc Ngoc

A thesis submitted in conformity with the requirements
for the degree of Master of Software Engineering

© Copyright by Nguyen Duc Ngoc 2021

**MINISTRY OF EDUCATION AND TRAINING
FPT UNIVERSITY**

Default Risk Prediction in Banking

by

Nguyen Duc Ngoc

A thesis submitted in conformity with the requirements
for the degree of Master of Software Engineering

Supervisor:
Dr. Vu Thu Diep

© Copyright by Nguyen Duc Ngoc 2021

Default Risk Prediction in Banking

Nguyen Duc Ngoc

Degree Master of Software Engineering

FPT University

2021

Abstract

The problem of predicting default risk or credit risk is a typical problem that is an important role in the financial and banking industry, this is a problem that has much researches and is still constantly improving. Early prediction of credit risks will help the banks and finance companies reduce risks and make the right decisions to increase profits. AutoML is a big step forward in information technology when the era of applying AI/ML advances widely in all industries, not only in finance or banking to help people reduce risks, save costs, increase productivity, and increase revenue. Although AutoML has many limitations, it is still being researched and developed every day. Applying AutoML gradually brings certain benefits to the AI/ML researcher and in the real business use case. In addition, AutoML is also a tool more accessible to employees in the financial and banking industry while they are not experts in AI/ML. Ultimately with the aim of making AI/ML more popular in the general public, using AutoML is a step in the right direction. We can consider AutoML as a revolution when using AI/ML in practice, and especially in the banking and finance industry. In this thesis, I describe specific steps on using H2O AutoML to create predict default risk models with Home Credit data and improve them in a simple and quite effective way.

Keywords— Default Risk, AI in Finance, AI in Banking, Model Interpretability, AutoML, Feature Explanation, H2O, AutoML

Acknowledgements

I would like to express my deepest gratitude to my instructor, Dr.Vu Thu Diep, who has the attitude and qualities of a teacher full of knowledge, wholeheartedly devoted to students, he has not stopped motivated me to finish my thesis. As an important person at the company, my work is always busy, without the enthusiastic guidance and encouragement of Dr.Vu Thu Diep, I would not have been able to complete the thesis.

I would like to express my deep gratitude to all the teachers who taught me at FSB in my master's program in software engineering. They have fully equipped me with the knowledge that is not only theoretical but completely applicable to my work, helping me to have a new role at FPT's AI unit - Quy Nhon AI. (QAI), an interesting and valuable job for the company as well as potential AI customers in Vietnam and many other countries.

In addition, I would also like to express my gratitude to a special person, my sister, MSc.Nguyen Thi Thuy Lien, a graduated student at FSB, has constantly helped me learn how to optimize and The in-depth theories needed on AI to complete the thesis. As a researcher at FPT Research Institute, she has excellently demonstrated her knowledge and experience as a researcher when guiding the latter.

Contents

List of Figures

1	Introduction	1
1.1	Motivation	1
1.2	Review	4
1.3	Research Purpose Statement	6
2	Fundamental Theory	7
2.1	Binary classification	7
2.2	Ensemble learning	7
2.3	Metrics	9
2.3.1	True/False and Positive/Negative	9
2.3.2	ROC, AUC and Gini	10
3	Approach Solution	11
3.1	Traditional approach solution	11
3.1.1	Understand the business	11
3.1.2	Collect the data	11
3.1.3	Prepare the data	12
3.1.4	Enrich the data	12
3.1.5	Predict	13
3.2	New approach solution - Auto Machine Learning	13
4	Experiment and discussion	17
4.1	Data Preprocessing	17
4.1.1	Data Understanding	17
4.1.2	Data Preprocessing	19
4.2	H2O Modeling and Performance	21
4.2.1	First experiment : Performance with original train data	21
4.2.2	Second experiment : Performance with combined all data files	27
4.3	Discussion	29

List of Figures

1.1	The History of Artificial Intelligence [2].	2
1.2	Artificial Intelligence vs Machine Learning [4].	3
2.1	Single ML model vs. Ensemble Learner	8
2.2	Boosting ensembler	8
2.3	Confusion matrix	9
2.4	ROC, AUC and Gini	10
3.1	Describe about traditional Machine Learning and AutoML	13
3.2	Traditional ML vs. AutoML SOURCE: JANAKIRAM MSV	14
4.1	Data Architecture Diagram	18
4.2	TARGET per	19
4.3	Distribution of Amount Credit	19
4.4	Import the train data	22
4.5	Parse the train data	23
4.6	Split the train data	24
4.7	Train the AutoML model - Config Parameters 1	25
4.8	Train the AutoML model - Config Parameters 2	25
4.9	Train the AutoML model - Training completed	26
4.10	Review the leader board	27
4.11	Train the XGBoost model	28
4.12	Train the XGBoost model - Training completed	28
4.13	View the XGBoost model	29
4.14	Variables Importances	30
4.15	Unnecessary features	30

Chapter 1

INTRODUCTION

1.1. Motivation

What is AI (Artificial Intelligence)? AI is defined as a simulation of the human brain that enables machines or systems to perform complex tasks that require intelligence. AI works like a brain, it can think and make highly accurate decisions based on the data provided. AI can also improve by learning and updating more data. The history of AI dates back to the 1950s when Alan Turing introduced the Turing Test to test the intelligence of computers. During the first 20 years from 1950 to 1970, there was not much progress because computers and systems were in their basic stages. It wasn't until the 80s when John Hopfield and David Rumelhart introduced deep learning techniques that allowed computers to learn based on the principles of biologically inspired neural networks, AI made remarkable strides. But by the 90s the research focus was shifted to enhancing computing power and augmenting with logic programming, so the research on AI has gradually declined [1]. The development speed of AI was only really accelerated after 2011 when technology giants such as Facebook, IBM, Microsoft, Google, Apple, Tesla, etc. began to participate in the development of applications using artificial intelligence. created to generate business benefits. Currently, AI is really booming and is applied in all fields to support people in their work from the medical industry, healthcare, automotive, logistics, finance, insurance... Everywhere or from any media channel we can hear the mention of popular phrases "The Fourth Industrial Revolution" and "The Artificial Intelligence". AI has been transforming many industries and services thanks to the power of data and the internet. The evolution of AI can be briefly described in the infographic of Fig 1.1.

What is ML (Machine Learning)? Definition Machine Learning is a branch of Artificial Intelligence that focuses on using statistical algorithms and data to learn on its own and improve its accuracy. ML is applied to solve many problems in many different fields, especially those that require data processing and interpretation [3]. ML is being applied to create advancements in search engines, self-driving cars, medical support, and especially in finance, banking, insurance... In fact, all Information Technology products and services on the market now very creative when applying ML, as recommendation tools help customers easily find the products and services that they need accurately and help sellers up selling in the most effective way. Tools to support remote medical examination and treatment based on images, help both doctors and patients save time, serve more customers, and help find diseases

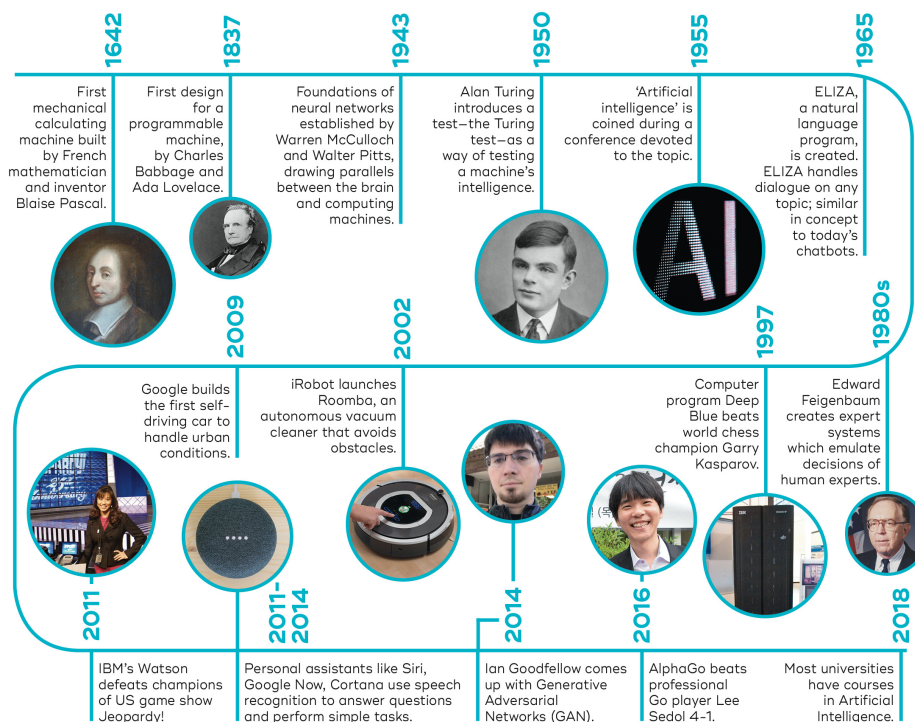


Figure 1.1: The History of Artificial Intelligence [2].

early and suggest treatment regimens. The treatment plan for mild diseases without the need to visit a doctor directly. Tools to prevent, early warning system against fraud, risk and support loan approval, as well as other tools to help fight money laundering, predict market trends in the financial, banking, and insurance industries. We can see ML's contribution in every field, but special in the financial industry, ML is used in the most diverse ways, from big data mining, information extraction from big data, data modeling, etc. data and create models to understand customer preferences in detail, manage risk, detect fraud, monitor behavior, automate customer support, and even verify identity (eKYC).

In the last 2 years, I have been involved in many projects on AI and ML. The first is the Deep Clinics project in the medical industry, supporting the examination and treatment of skin diseases such as acne, skin fungus, skin cancer, customers can use mobile applications or websites to take photos and send photos to the disease classification model and give a treatment regimen for middle diseases, with high levels will be recommended to meet a doctor directly. The second project is a project to build an application that automatically extracts information according to the form from the invoice images and tables submitted by users to the system, users can easily create templates of invoices and tables. The form then uploads the images, the system will automatically recognize the data in the marked areas and input it to the data system. The third project is TPBank's Chatbot project to support corporate customers, which is a Chatbot serving corporate customers with the functions of consulting and providing services exclusively for TPBank's corporate customers. The fourth

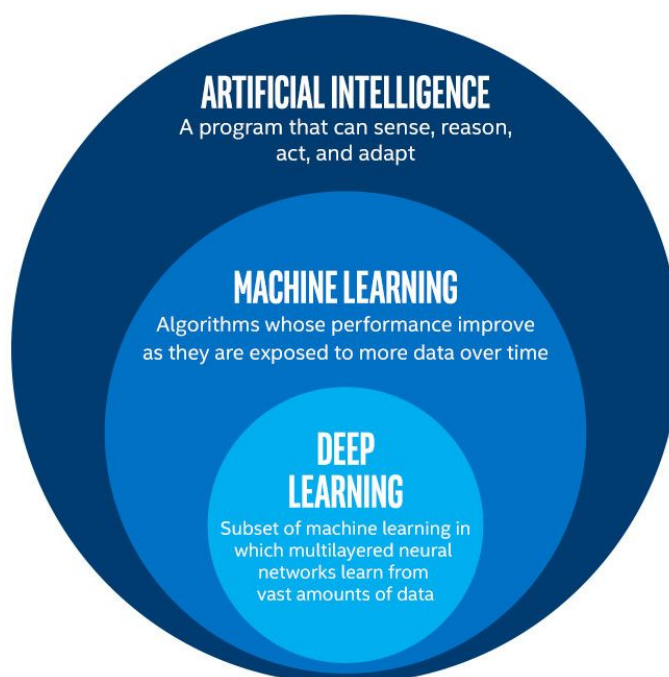


Figure 1.2: Artificial Intelligence vs Machine Learning [4].

project is building ML models to support Risk Management, Credit Management Center, at TPBank. The fifth project is a national project on building an evacuation simulation tool and applying ML to calculate escape time in construction works for the Fire Prevention and Fighting Department of the Ministry of Public Security. In the future, I will join many other AI and ML projects that are in the processing or being proposed to customers, especially the banking domain. So, even though I am a participant in many AI projects in different domains, but based on the actual needs and trends of my work, I decided to choose the thesis topic of Default Risk (Credit Risk) in Banking.

In the financial industry, the possibility of a lender experiencing a loss due to the borrower's inability to repay is known as Credit Risk. In these cases, the lender faces the risk of default on each extension or renewal of credit. Imagine financial credit institutions that have thousands of credit lines for organizations and individuals such as TPBank, MaritimeBank, VPBank... The obvious luck here is the probability of default of a person or a person. the organization is not random. Which can be calculated and predicted by external factors such as occupation, salary, living area... This is where financial analysts can intervene to find measures to avoid risks. to lending agencies. For this thesis topic Default Risk, I will use data on Kaggle because no agreement has been reached with TPBank on the use of data. In the future, after completing the thesis, I will continue to ask TPBank to also provide data set so that I can publish an article and propose a contract to build a Default Risk model for TPBank's RM department.

1.2. Review

In general, when applying AI/ML to the banking and finance industry, banks can get five main benefits as Increase revenue (Marketing), Avoiding risk of non-compliance (Risk), Decreased costs (Operations), Competitive edge (Innovation), Speed to value and team efficiency (Operation).[5] For Increase revenue, we can mention applications such as Revenue Attribution that allocates revenue to branches, AI systems will rely on data to provide an optimal solution to allocate customers to branches to ensure the best service for customers as well as ensure revenue evenly distributed to branches. Customer Churn helps predict customers may leave the bank based on customer activity data, recently additional support systems such as voice analysis via recording, voice call, or video analysis have also appeared. Photo-based on-camera recorded at the counter. Recommending and Up selling is a cross-selling product recommendation system to increase sales based on the analysis of data collected from customer habits, to perform this function, a large amount of data is required. about customer behavior. Insurance banks or insurance financial institutions can benefit from building AI/ML application systems, observing market fluctuations, adjusting prices according to fluctuations or automated systems. Approving insurance levels for customers will help customers feel comfortable and reduce risks for insurance organizations. For Reduce Risk, we can mention applications such as Cyber security, every year banks are subjected to large attacks from vulnerabilities, the application of AI/ML systems can help find vulnerabilities. security vulnerabilities automatically, or detect anomalies in log files. Fraud Detection and Prevention is an AI/ML application to handle contextual or collective unusual data, such as detecting abnormal withdrawals from credit cards for early warning or checking signatures. AML, KYC, and Wire Transfer Fraud is a system that uses AI/ML to automatically identify customers and check international money laundering transactions by cross-comparison between AML databases and anomalies in remittance transactions. Trade Failure Prediction is the application of AI/ML to learn data about transactions so that it can predict future transactions that will fail and can find the cause of the transaction failure, so that Fix these errors to increase customer satisfaction as well as make the system work better. Credit Risk & Loss Forecasting is the application of this thesis, this is an application that uses AI/ML to predict credit risk based on data collected from many sources such as credit agencies. government or independent organizations. Cash Management Product Risk is a basic and essential application in banks, it is an application that uses AI/ML to quickly identify risky invoices, duplicate payments, etc.

For Decreased costs/speed to value & team efficiency, which can include Improvement and Automation of Processes applications, banks are now applying robotics automation processes (RPA) to automate repetitive tasks, it can save a lot of resources and time, especially during the COVID19 period, all banks need to reduce costs and increase production efficiency. At TPBank2019 is the year of RPA when all processes have been integrated with FPT's RPA Ak-

abot tool. In addition, automation also helps to quickly and conveniently extract information from a data mart, and digitize processes is also easier. Replacing Any Rules-Based System is the replacement of traditional rule-based sets with new AI/ML models, currently, this work cannot be completely replaced but can be combined using old rule-based with other AI/ML model in the new system and then when there is data for retraining, banks can completely replace the AI/ML model for rules-based. At TPBank I once proposed an AI/ML model system System integration allows configuring newly added rules and adjusting rules-based to detect card expiration transactions or increase card limit abnormal. Streamlining MRM, MRM is a sub-branch of RM, which streamlines the way models are built and deployed. MRM requires a consistent system or process by which models are delivered so that the steps from AI/ML model building to production adoption can be completed seamlessly and risk-free. Regulatory Reporting Automation is automatic reporting according to regulations. Previously, the synthesis of reports was very complicated based on aggregated data files, but now with the application of AI/ML, banks can completely build reporting and prediction systems proactively and quickly.

For the competitive edge, there are applications such as Analyzing Money Flows, Competitive and Benchmark Analysis, this is the use of AI/ML models to predict market factors and bank competitiveness based on different sources, we are currently being asked by TPBank to build a system including an AI/ML model that uses TPBank's own market tracking data to analyze factors influence of the CPI index, which is a system that is considered separate from actual data tracked over many years rather than using time series data provided by other organizations due to concerns being makeup when submitting the reporting. Evidence-Based Research is the use of valuable data in practice to research and build models that deliver competitive performance against competitors. Alternative Dataset Valuation in Research is the selection of data sets in research, which can use data sources from market research companies as well as real data stored and tracked for many years by the government. For banks, this is valuable real data, so the research and construction of the model will be closer to reality and bring more competitive efficiency.

Above we have gone through each use case for departments of banking institutions, next we will go into the application of Credit Risk in banking. Credit Risk or Default Risk, in general, is the application of classification methods based on the past data of customers to analyze the possibility of overpayment in the future. Some techniques commonly used in credit risk scoring include discriminant analysis, logistic regression, Bayes classifier, nearest neighbor, artificial neural networks, and classification trees. Artificial neural networks are said to be more accurate than the other methods. Methods and models are being developed and continuously updated to address the key issue at the bank of accurate customer classification and credit risk prediction. Based on these increasingly accurate results, banks can increase their loans and thus make larger profits. Galindo and Tamayo (2000) research, through statistical engineering analysis and machine learning, built more than 9000 models as part

of the study and ranked the performance of various algorithms. They show that the CART decision tree gives the best estimate with default risk and the Neural Network comes in second. Hamori et al. (2018) studied and compared the accuracy of prediction and classification with bagging, random forest, augmented with neural network methods in default payment data analysis. They found boosting's ability to outperform any of the studied machine learning methods. Several researchers have also evaluated the application of hybrid techniques and synthetic methods to credit scoring studies (Bastos 2014; Hamori et al. 2018; Raei et al. 2016). "Support Vector Machine" (SVM) is a supervised machine-learning algorithm and is widely used in classification problems. The SVM algorithm is applied either natively or in some form to build credit scoring models (Bellotti and Crook 2009; Cao et al. 2013; Van Gestel et al. 2003; Huang et al. 2007; Lai et al. 2006)[3]. Thus, Neural networks, Support Vector Models, and Random Forest are the most popular algorithms used in credit scoring, there are not many studies and articles related to AutoML and default risk prediction. And I decided to use AutoML to experiment with the default risk in this thesis.

1.3. Research Purpose Statement

For this study, it is necessary to achieve the following two research objectives:

- The current status of AI in the Banking industry market is still a large unexploited business segment, all banks are looking to apply AI to banks to support operations and improve customer service quality. Normally, companies in Vietnam when implementing a project, require software companies to provide PoC (Proof of Concept) within 2 weeks to decide whether to deliver the package to that software company or not. So as a person in charge of AI in Banking at QAI - FPT Software, and implementing many AI projects in the banking industry in Vietnam and many other related fields, I decided to delve into research. AutoML with H2O to quickly implement PoCs for customers upon request, specifically, this Default Risk topic will be proposed to the Risk Management department at TPBank.
- Currently, TPBank is looking to build a lot of AI models, in addition to hiring software consultants to help build AI models, TPBank also wants to provide an AI training support package for bankers. The Risk Management department alone needs 40 different models, and there are only 4-5 data scientists working on model building. From that fact, I realized that it is necessary to find a simple, fast and effective method to build a model for employees who have almost no deep knowledge of AI that can be easily familiarized and applied to their work quickly. For bankers, model building is still a new job that requires a lot of knowledge that they lack, so the more effective, simple, and better the model-building method, the less time the better. At TPBank, there are a number of models that have been applied in practice, TPBank's requirement is that the models built must have a GINI greater or equal to 0.5, equivalent to an AUC greater or equal to 0.75.

Chapter 2

FUNDAMENTAL THEORY

In this chapter, some of the background theories related to the techniques applied in later chapters will be repeated.

2.1. Binary classification

Binary classification problems [6] is the task of classifying the elements of a set into two categories. Such as the problem covered in this thesis is to classify customer credit applications into "group at risk of default" and "group with no risk of default". Popular algorithms to deal with binary classification include:

- Logistic Regression
- k-Nearest Neighbors
- Decision Trees
- Support Vector Machine
- Naive Bayes

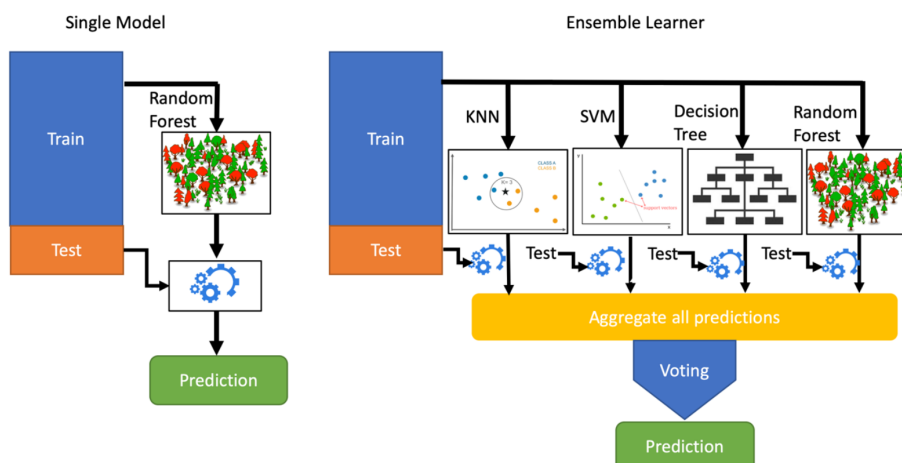
2.2. Ensemble learning

Ensemble learning is the process by which multiple models, such as classifiers or experts, are strategically generated and combined to solve a particular computational intelligence problem[7].

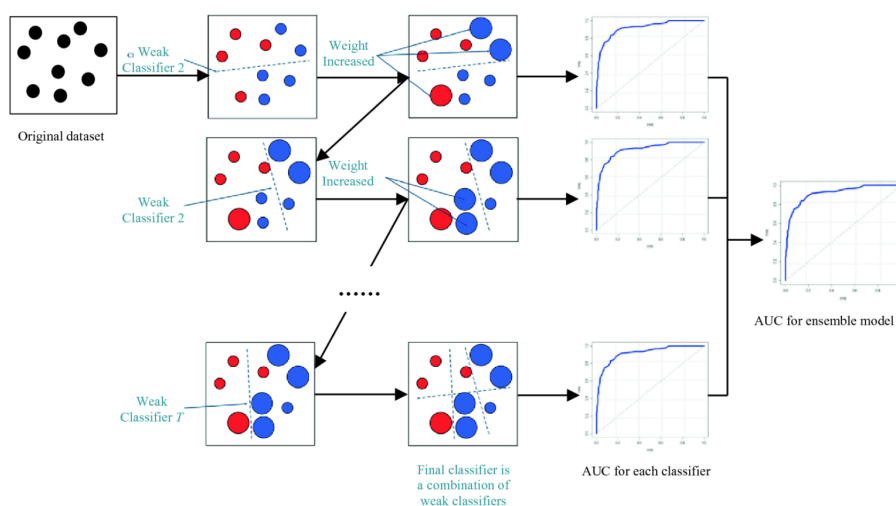
There are some commonly used ensemble learning techniques:

- Simple: such as Max Voting, Averaging, Weighted Averaging.
- Advanced: such as Stacking, Blending, Bagging, Boosting.

Bagging, which stands for bootstrap aggregating, is one of the earliest, most intuitive and perhaps the simplest ensemble based algorithms[8]. In the first step of bagging techniques, engineers have to create some sub-sample datasets by randomly drawn – with replacement – from the entire training dataset. Then build a large number of models on these different sub-samples. These models will be trained independently and in parallel. They are then combined by a simple majority vote for their decisions.

Figure 2.1: Single ML model vs. Ensemble Learner

Boosting: Similar to bagging, developers have to build a large number of models. Each of the following models will learn to correct the errors of the previous model, or data that the previous model predicts incorrectly. Therefore, it forms a series of models where the latter model will be better than the previous model because the weights are updated through each model by the way that the weight of the correct data will stay the same, but the weight of the wrong data will be increased.

Figure 2.2: Boosting ensemble

Gradient Boosting Machines (GBMs) is a well-known algorithm used to solve regression and classification problems. Instead of gradually increasing the weights of the wrongly learned samples, in Gradient Boosting the following learners will learn the error of the previous learner.

XGBoost, stands for Extreme Gradient Boosting, is an open-source software library which provides an optimized distributed gradient boosting framework for various programming language such as C++, Java, Python, R, Julia, Perl, and Scala. XGBoost is rated as highly

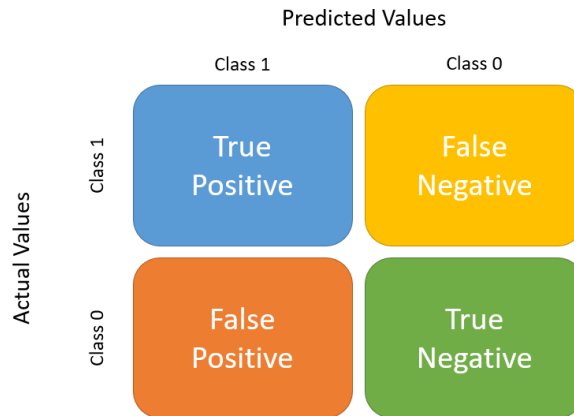
efficient, flexible and portable. XGBoost provides a parallel tree boosting (also known as GBDT, GBM) that solve many data science problems in a fast and accurate way[9].

2.3. Metrics

2.3.1 True/False and Positive/Negative

In the binary classification problem, the true positive is the result that the model correctly predicts the positive class, and the true negative is actually the result that the model correctly predicts the negative class. Similarly, a false positive is an outcome where the model incorrectly predicts the positive class. And false negative is the result that the model incorrectly predicts about the negative class.

Figure 2.3: Confusion matrix



TPR (True Positive Rate / Sensitivity / Recall): Represents the percentage of correctly classifying positive samples overall positive samples, calculated by the formula:

$$TPR = \frac{TP}{TP + FN}$$

The higher the TPR, the more accurately positive samples are classified. Specificity: represents the correct classification rate of negative samples on all the negative sample, is calculated by this formula:

$$Specificity = \frac{TN}{TN + FP}$$

FPR (False Positive Rate / Fall-out): is representation of the rate of mislabeling negative to positive samples across all negative samples, calculated by the formula:

$$FPR = 1 - Specificity = \frac{FP}{TN + FP}$$

It can be seen that Specificity is inversely proportional to FPR. The higher the FPR, the lower the Specificity and the greater the number of mislabeled negative samples.

2.3.2 ROC, AUC and Gini

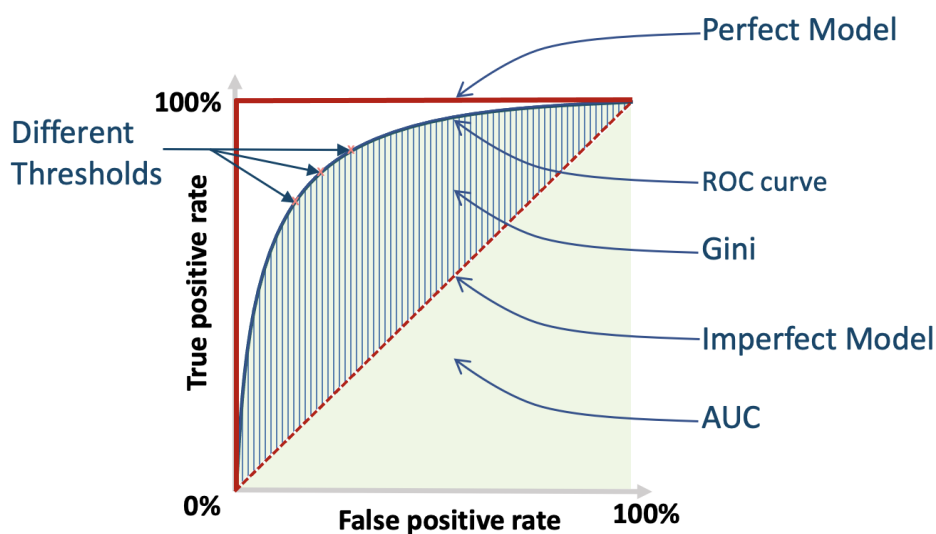
A ROC (receiver operating characteristic) curve is a graph showing the performance of a classification model at all classification thresholds [x]. This curve plots two metrics TPR on the y-axis and FPR on x-axis.

AUC (Area Under the ROC Curve): The ROC curve itself is not a metric as it is just a curve that we want to see in every threshold TPR and FPR metric relative to each other. Thus, a clearer metric was proposed to quantify this curve and compare the two models. It is AUC, which stands for area under the ROC curve. With this name, obviously it measures the entire two-dimensional underneath the ROC curve. The closer the AUC index is to 1, the more accurate the model is. The closer the AUC is to 0.5, the worse the classification performance, and if it is close to 0, the model will reverse-classify the results (classifying positive to negative and vice versa).

The Gini coefficient is just one way of adjusting the AUC, to make it more understandable and meaningful. The value will now be in the range [-1, 1]. It is more natural to look at a random classification model with a representative number of 0, and -1 representing the inverse classifier model.

$$Gini = 2 * AUC - 1$$

Figure 2.4: ROC, AUC and Gini



Chapter 3

APPROACH SOLUTION

In this chapter, I will recall some concepts related to the traditional approach to solving the default risk prediction problem, as well as the approach using AutoML.

3.1. Traditional approach solution

Essentially the analysts will be responsible for assessing this risk by analyzing the borrower's ability to pay off debt. But now, with the development of machine learning, the work amount of credit analysts has been greatly reduced. Machine learning algorithms have provided the world with powerful credit risk assessment tools with predictive speeds that are far superior to traditional analytical methods. Although there are different methods for dealing with individual problems, we can generalize, there are some fundamental steps to complete a data analysing project in the traditional approach solution:

3.1.1 Understand the business

This step is the key to ensuring project success and is also the first phase of any data analysing project. Even if it is a personal project or a research project devoted to learning algorithms, this step is still necessary. To have motivation, direction, and purpose, we still have to define what we need to do with the data. In this step, the data analyst also needs to build an overview of the path to achieving the desired results.

3.1.2 Collect the data

There are many ways to collect data, depending on the purpose of the project. Retrieve from existing databases: This is an easy way to get data for people who already have an understanding of how the organization works. However, if you don't work for that organization, the request for data will be quite difficult to accept.

(a) Crawl Data from a website: There are many ways to crawl data from a website that we can use. (1) Use Website APIs, since many large social media websites, like Facebook, Twitter, Instagram, StackOverflow provide APIs for users to access their data. However, data APIs are not available for all the websites, some of them don't provide any public API. The reason

may be due to technical limitations or just simply that they don't want to be in charge on that matter. In this case, we can use (2) ready-to-use crawler tools such as Octoparse or import.io.

(b) Look for open data sources: Along with the development of the internet, and the strength of the researchers community, the data sets are widely shared. We have enough data sets to enrich what we have with additional information. For example, census data will help data analysts add the average revenue for the county where the user lives.

(c) An extremely important factor that cannot be overlooked when collecting data is to ensure that the data collected complies with data privacy regulations. Privacy and protection of personal data is becoming a priority for users, organizations and legislators.

In this thesis, I used an open data source provided by a well-known financial company in the hope of being able to build a general data analyzing and processing that could be used in any organization which has a similar business.

3.1.3 Prepare the data

Once data analysts have the data they want, the next step will be the data processing. This is the process that takes up about 80% of a data project's time. In this step, analysts will begin to dig deep to find the characteristics of the data and find ways to link them together to achieve the original goal.

In this process, data cleaning is considered the longest and most annoying step. Let's imagine, despite having national geography data, you have different spellings or even missing data. And such ambiguous data cannot be used directly in data projects. This is the time when data analysts must review each column to ensure the data is consistent and clean.

3.1.4 Enrich the data

Once the data is clean, now the analysts are ready to process it. How to get the most out of the data? This data enrichment phase can begin by combining all the sources to narrow the data down to the necessary features. Another way to enrich data is to combine data sets.

When dealing with large volumes of data, visualization is the best way to discover and transfer the analyst's findings. Graphs are also another way to enrich your data set and develop more interesting features.

When collecting, preparing, and processing data, data analysts must take great care not to insert unwanted bias or other undesirable patterns in it. One of the things that scare people about data and AI is that algorithms can't recognize bias.

As a result, when training the model on biased data, it understands the bias as a reproducible decision and not something to correct. This is why an important part of the data manipulation process is to ensure that the used data sets do not reproduce or enforce any biases that could lead to skewed or unfair results.

3.1.5 Predict

At this step, machine learning algorithms can help analysts go a step further in better understanding and predicting future trends. Clustering (aka unsupervised) algorithms can help build models to uncover trends in data that are indistinguishable in graphs and statistics. These events produce groups (or clusters) of similar events and more or less clearly show which characteristic is the determining factor in these outcomes.

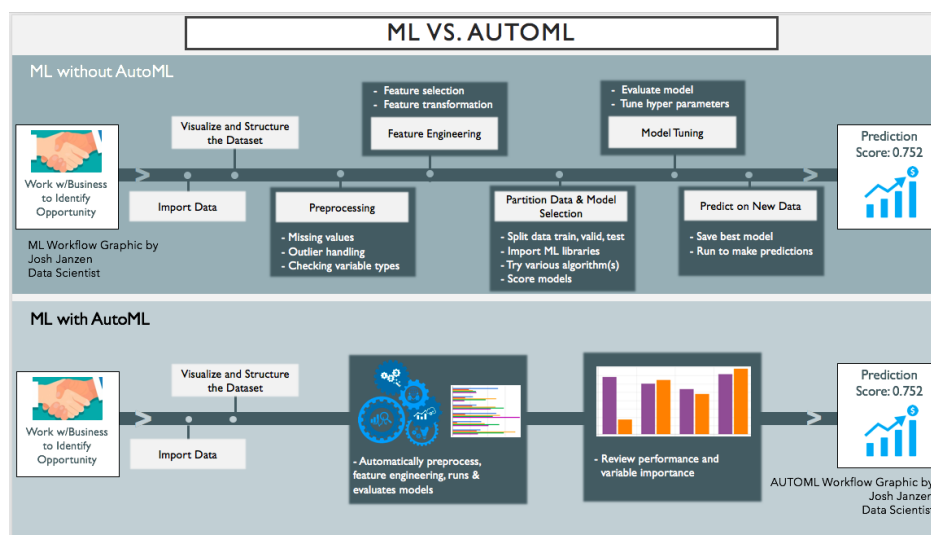
On the other hand, with supervised algorithms, data scientists can go even further and predict future trends. Analyzing the past data, they find characteristics that have impacted trends in the past, then use them to build predictions in the future. More than just acquiring knowledge, this final step can lead to building entirely new products and processes.

Machine learning models will actually degrade over time if they are not continuously improved and provided with new data. Therefore for models to remain useful and accurate, data scientists need to constantly reevaluate, retrain, and develop new features. This process needs to be iterative.

3.2. New approach solution - Auto Machine Learning

In recent years, the field of machine learning has grown and attracted many people to join. Even so, the demand for machine learning specialists still outstrips supply. That spurred research and found a tool for non-professionals to get into the field with relative ease, which is where AutoML, which stands for Automatic Machine Learning, was born.

Figure 3.1: Describe about traditional Machine Learning and AutoML



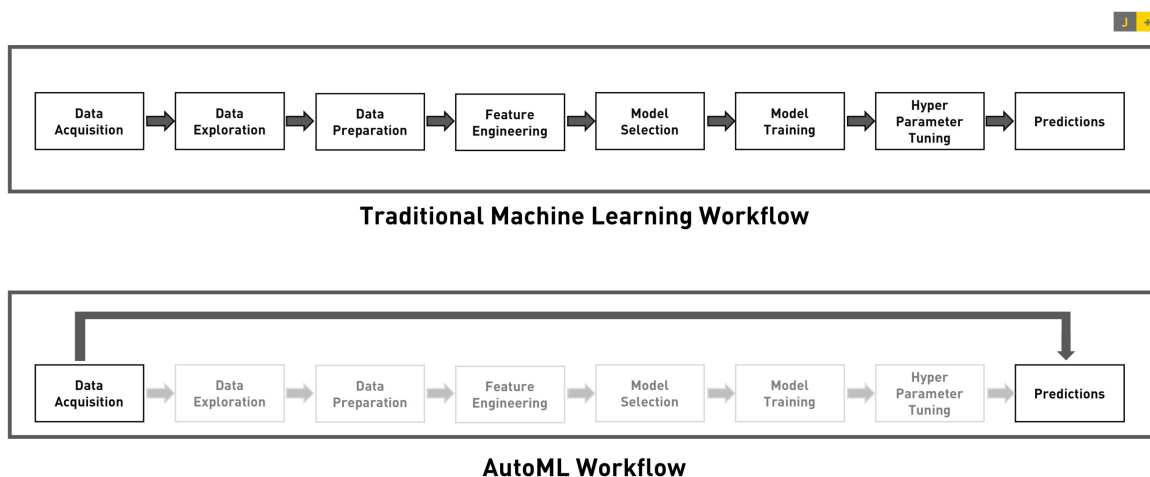
The goal of AutoML is to enable people with limited machine learning background knowledge to use the machine learning models easily[10]. With AutoML, data scientists will be able to synthesize machine learning solutions without having to go through a series of methods of

data preparation and refinement, model selection, model hyperparameters generation and so on.

This is a quick "cheat" way to achieve high accuracy for machine learning tasks without much effort. With AutoML, machine learning becomes more accessible, requires less expertise, and improves the overall performance of models.

The two main tasks of AutoML are data collection and prediction. In particular, these AutoML tools mainly support hyperparameter editing, iterative modeling, algorithm selection, model evaluation, as well as many other tasks.

Figure 3.2: Traditional ML vs. AutoML SOURCE: JANAKIRAM MSV



Here are some useful AutoML tools for automation:

- **Amazon Lex** Amazon Lex is the Deep Learning technology used in Amazon Alexa. It is an efficient service for building conversational interfaces in applications with voice and text. Amazon Lex provides the advanced deep learning functionalities of automatic speech recognition (ASR) for converting speech to text, and natural language understanding (NLU) to recognize the intent of the text, to enable you to build applications with highly engaging user experiences and lifelike conversational interactions[11].

Built close to AWS, Amazon Lex also offers built-in integrations with AWS Lambda, AWS MobileHub, and Amazon CloudWatch. In addition, users can also easily integrate AWS services such as Amazon Cognito and Amazon DynamoDB.

- **TPOT**

TPOT, which stands for Tree-based Pipeline Optimization Tool, was developed in the Computational Genetics Lab at the University of Pennsylvania. TPOT is one of the very first AutoML methods and open-source software packages developed for the data science community[12].

TPOT relies on the scikit-learn library, and strictly adheres to its API. It can also be used in both regression and classification tasks, and has special applications in medical research.

The developers call TPOT a genetic search algorithm, with the ability to find the best parameters and models[13]. In which, TPOT automatically finds and optimizes an ML pipeline includes of features preprocessing, model selection, and training, while most AutoML tools pay attention to model selection and model learning only. TPOT developers admit that it can be quite time-consuming to run[13].

- Auto-scklearn

Auto-scklearn is an automated machine learning toolkit and a drop-in replacement for a scikit-learn estimator[14]. it is a great result of researchers at the University of Freiburg.

Built on the scikit-learn library, auto-scklearn is capable of automatically searching the learning algorithm for new datasets, as well as optimizing their parameters.

Auto-scklearn provides highly innovative supervised Machine Learning where pre-processing is divided into data preprocessing and feature preprocessing. Instead of focusing on finding neural architectures for deep neural networks, Auto-Scklearn uses Bayesian optimization for hyperparameters, to be compatible with “traditional” machine learning algorithms in scikit-learn.

- Auto-Keras

Auto-Keras is an opensource AutoML system based on Keras, developed by DATA Lab at Texas A&M University[10]. The system runs in parallel on CPU and GPU, with an adaptive search strategy for different GPU memory limits.

Auto-Keras also utilizes the Neural Architecture Search (NAS) algorithm but applies “network morphism”. Network morphism is one of the paradigms to construct deeper neural networks on existing ones by morphing a well-trained neural network into a new one with the network function completely preserved[15].

Auto-Keras is easy to use, provides lot of inbuilt thing like cov2d, conv2D, maxPooling layers, and many data processing libraries. Addition, it also provides functionality to develop models on mobile device.

- H2O.ai

H2O is an open source, in-memory, distributed, fast, and scalable machine learning and predictive analytics platform that allows you to build machine learning models on big data and provides easy productionalization of those models in an enterprise environment [16].

H2O AutoML helps automate Machine Learning in various ways, from training to adjusting hyperparameters in the model. The AutoML process can be controlled by setting a specific performance or time limit, through which AutoML will return a table summary of the best models.

The H2O's main code is written in Java, thus allowing the framework to perform multi-threaded tasks. In addition, to improve efficiency, H2O integrates suitable interfaces for R, Python, and several other programming languages.

Because these reasons: (1) H2O can be accessed from R Studio, not only JAVA etc, (2) H2O provide the clickable H2O Flow interface (WEB UI) extremely easy to use, and (3) it can be used as open-source, I chose H2O for this thesis. The techniques used in this thesis will be described in detail in the following chapters.

Chapter 4

EXPERIMENT AND DISCUSSION

In this chapter, the data processing, as well as the AutoML H2O techniques that be used to solve the Default Risk Prediction in Banking problem, are described in detail.

4.1. Data Preprocessing

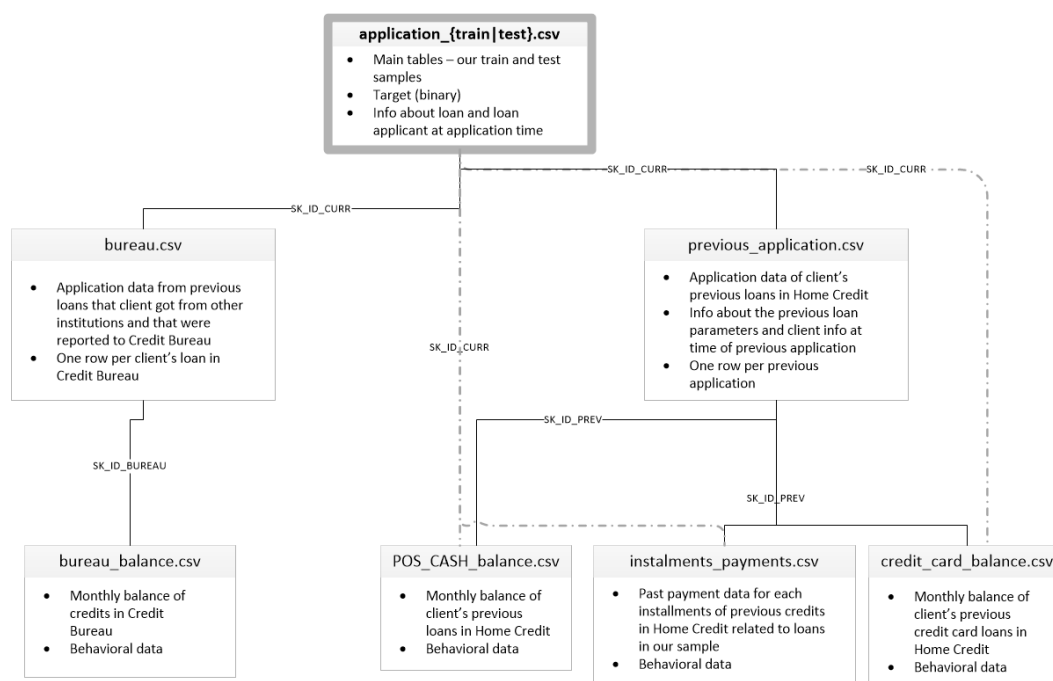
4.1.1 Data Understanding

Home Credit is one of the leading international companies in the consumer loan and installment field with outstanding advantages: fast, convenient, and friendly. Home Credit's goal is to increase the number of customers, which has helped the company improve its revenue. However, the more loans, the riskier it is. The large and diversified customer base increases the probability that Home Credit encounters borrowers that have problems in loan repayment. Thus determining the ability of customers to default becomes the most important task of Home Credit. How to correctly provide loans to individuals who are able to repay and reject unable customers?

The data-set that I used for this thesis is provided by Home Credit from several data sources. It consists of 10 files in the CSV format. The interrelationships between these files are depicted in Figure 4.1.

There are two main tables, divided into train and test set, corresponding to `application_train.csv` file and `application_test.csv` file. These two files are static data for all applications, each row represents one loan in the data sample. In easier words, this is the main information for personal loans, and it contains a significant amount of useful information for predicting credit default also. Other files contain a lot of information regarding previous applications, credit card balance, repayment history and so on. This is a vastness of information that promises a great deal of value in mining.

The training set holds 307,511 observations of 122 variables, each observation is the information of a person who received loans. In which, the "TARGET" indicates whether clients have difficulties in meeting payment or not. In this thesis, I considered this variable as default status. However, it can not be said that anyone who has difficulties in dept payment has actually defaulted. The remaining 121 features include the demographic variables, and some other information about the loan. These features can be classified into 5 categories,

Figure 4.1: Data Architecture Diagram

including: personal information, loan information, information about the customer's living area, documents that the customer provides and requests to send to the Credit Bureau.

The test set contains 48,744 samples, but unfortunately the "TARGET" column is not available because it was kept private for the Kaggle competition. Thus I did not use the test set in this thesis scope.

Credit bureau, is a data collection agency that gathers account information from various creditors and provides that information to a consumer reporting agency. The bureau.csv file contains all of the customer's previous credits provided by other financial institutions that have been reported to the Credit Bureau. Not every customer has a credit, and a customer can have multiple credits, the number of rows with the same customer identifier is the number of credits the customer had in the Credit Department before the application date.

The bureau_balance.csv file contains monthly balances information of previous credits in Credit Bureau. Each row corresponds to a month of credit history stored in the Credit Bureau. So the number of rows of this table will be equal to the number of customers * the relative previous credits * the number of months where history be observed for the previous credits.

POS_CASH_balance.csv file contains monthly balance snapshots of previous POS (point of sales) and cash loans that the applicant had with Home Credit. Similarly, the credit_card_balance.csv contains balance information of previous credit cards.

If clients have any previous application for Home Credit loan, information of these applications should be saved in the previous_application.csv file. Each row matches a previous application related to loans.

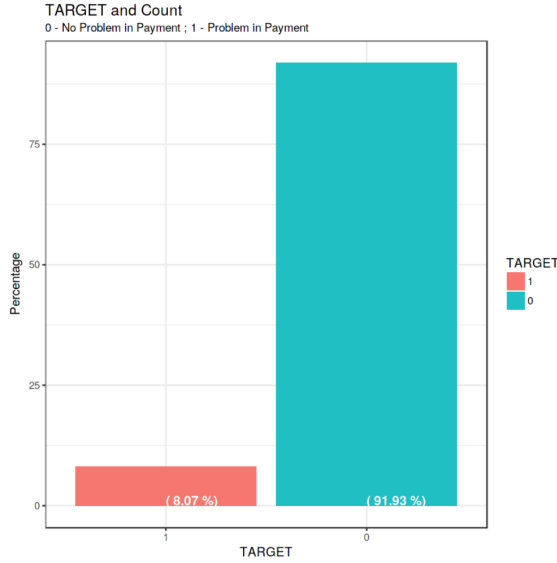


Figure 4.2: TARGET per

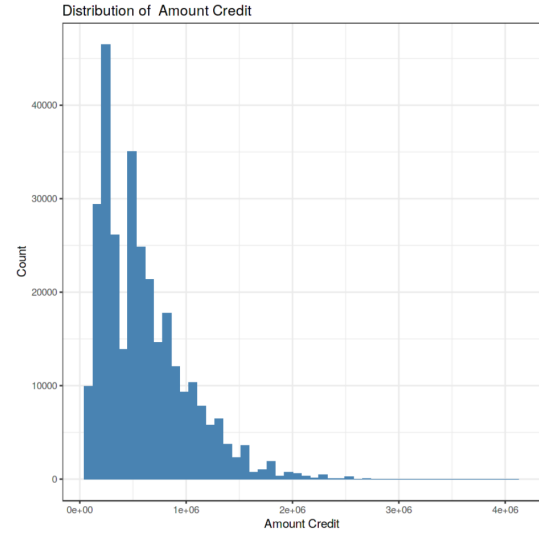


Figure 4.3: Distribution of Amount Credit

Finally, the repayment history for credits disbursed at Home Credit will be stored in the `installments_payments.csv` file. One row corresponds to one payment of one installment or one installment equals one payment of one previous credit at Home Credit.

4.1.2 Data Preprocessing

The samples in Home Credit dataset are divided into 2 classes: "TARGET of 1" - have problems with payment, and "TARGET of 0" - no problem with payment. The figure 4.2 indicates that Home Credit is an imbalance dataset, the number of samples with a TARGET of 0 accounted for 91.93%, much larger than samples with TARGET of 1.

The figure 4.3 below shows the the distribution of amount of credit applications in the dataset. Looking at it, can see that loans concentrated in the range of 200 700 thousand dollars.

I have built two experiments within this thesis. In the first experiment, only file `application_train.csv` was used. In the second experiment, I combined all the data files into a unified data block to put into training.

4.1.2.1 First experiment

In the first experiment, I just used `application_train` only. The reason is that this information very clearly affects the final prediction results. The dataset will be divided into 2 parts at a ratio of 0.75:0.25 corresponding to training data, and testing dataset which used to compare the final results.

4.1.2.2 Second experiment

As mentioned above, the `application_train` table has each row corresponding to a loan application from a customer. However, one `SK_ID_CURR` (customer identifier) can exist in multiple rows in other tables. The function `summarise_all()` is used to summary all variables of a table by group. And then the the funtion `left_join()` is called to merge two or more tables together.

The code below is executed for the purpose of: (1)Summarizing all variables of `bureau_balance` table by group which has the same `SK_ID_BUREAU`. (2)Merging the information from `bureau_balance` to `bureau` table.

```
fn <- funs(mean, sd, sum, .args = list(na.rm = TRUE))

sum_bureau_balance <- bureau_balance %>%
  mutate_if(is.character, funs(factor(.) %>% as.integer)) %>%
  group_by(SK_ID_BUREAU) %>%
  summarise_all(fn)
rm(bureau_balance); gc()

sum_bureau <- bureau %>%
  left_join(sum_bureau_balance, by = "SK_ID_BUREAU") %>%
  select(-SK_ID_BUREAU) %>%
  mutate_if(is.character, funs(factor(.) %>% as.integer)) %>%
  group_by(SK_ID_CURR) %>%
  summarise_all(fn)
rm(bureau, sum_bureau_balance); gc()
```

Handling similar to the remaining data tables. Finally merge all the information into a single table as the following code:

```
train_test <- train %>%
  select(-TARGET) %>%
  bind_rows(test) %>%
  left_join(sum_bureau, by = "SK_ID_CURR") %>%
  left_join(sum_cc_balance, by = "SK_ID_CURR") %>%
  left_join(sum_payments, by = "SK_ID_CURR") %>%
  left_join(sum_pc_balance, by = "SK_ID_CURR") %>%
  left_join(sum_prev, by = "SK_ID_CURR") %>%
  select(-SK_ID_CURR) %>%
  data.matrix()
tri <- 1:nrow(train)
```

```
y <- train$TARGET
```

The Home Credit dataset contains a lot of missing values. To handle these cases, I fill in the missing slots with median value of that column. This is easily handled in R with the following simple statement:

```
for (col in missing_values$col_name){
  train_test[is.na(train_test[,col]),col] <- median(train_
    test[,col], na.rm = TRUE)
}
```

Once the data processing is done, I will write all the combined data to a new CSV file. This step will be very helpful to convert data to H2O Flow. Note that we can completely use H2O on R Studio, but for the scope of this thesis I will mostly use H2O Flow.

```
# Write new data to csv file
write.csv(train_test, "../H2O_Data/train_all_features.csv",
  row.names = FALSE)
```

4.2. H2O Modeling and Performance

4.2.1 First experiment : Performance with original train data

For the first experiment, I will try to use only application_train.csv file from train data folder. This file contain data already label which customer is default. I will use R for programming and H2O Flow for training, checking leader board and generate model.

At the first time we need import all the library needed, then start the H2O virtual machine in R Studio. After that we move to browser and access to <http://127.0.0.1:54321/flow/index.html> for H2O Flow UI. At the Flow UI we will import the train data, parse train data, split train data and training the model. H2O Flow is an H2O user interface that allows us to interact with H2O, using functions such as file import, data integration, data decomposition, model training, model improvement, and retraining. , and time provides model downloads. H2O Flow also supports R scripts, Rest API, and CoffeeScript, but we don't need to know the code when using H2O Flow, which is quite convenient for bank staff when training models. H2O is sequences of executable cells, which are built-in functions with a specific function that helps you access functions or data objects. We can easily save Flows as files and reload them for the purpose of storing or transferring model-building files.

```
library(tidyverse)
library(data.table)
library(caret)
```

```
library(h2o)
```

```
h2o.init(min_mem_size = "2G")
```

```
h2o.no_progress()
```

```
h2o.removeAll() # Clean slate in case already has data, model
```

4.2.1.1 Step 1 : Import the train data

We browse to the directory contains all data files and select the `application_train.csv` file Search box, then click to Import button, all the data will import shortly to H2O.

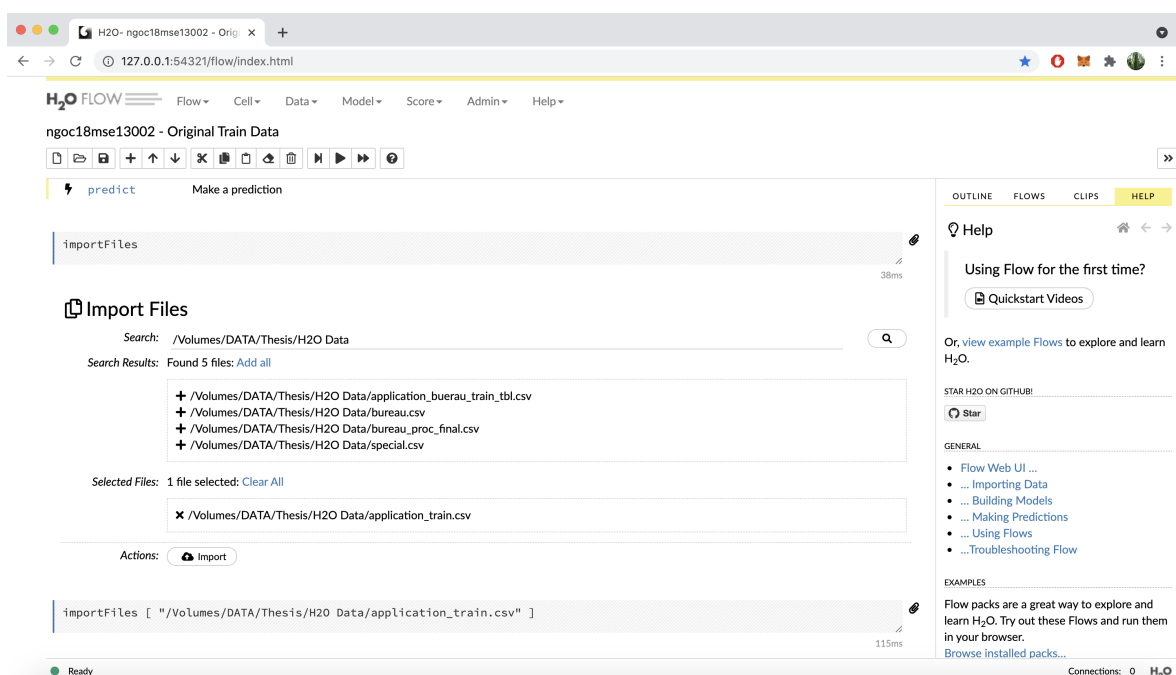


Figure 4.4: Import the train data

4.2.1.2 Step 2 : Parse the train data

After import the data, we can parse data to the H2O data frame. H2O provides some features to in parse data function such as:

- Parse type as AUTO, ARFF, XLS (BIFF 8 only), XLSX (BIFF 8 only), CSV, SVMLight, ORC, AVRO, PARQUET files.
- Select a separator if a separator or delimiter is used.
- Select a column header option as Auto, First row contains column names and First row contains data.

- Select any necessary additional options as Enable single quotes as a field quotation character and Delete on done.

At the *TARGET* column, it's labeled for default, so we must change it to ENUM type. There are several types in train data as ENUM, NUMERIC, and STRING. We can pre-process data as encode categorical variables from R language, but with AutoML we don't need to do that, H2O will processing it automatically. We can preprocess data on H2O Flow but for this data preprocessing, H2O Flow is not strong. H2O Flow only provides some available interfaces to impute data by column but does not allow multi-column processing, for data sets with too many features, the processing will be very complicated. So for the data preprocessing I will do it on R Studio, and provide the post-processing to H2O Flow.

Figure 4.5: Parse the train data

4.2.1.3 Step 3 : Split the train data

After parsing the data, we got an H2O data frame of all train data with labels as *TARGET* column, we must device train data to two sets, train set for training model and test set for valid the model. We can use the ratio of 0.75/0.25. This train data was label by Home Credit, so with the test set as 0.25 without the *TARGET* column we can use it as valid data, the test set now can be ground truth data set. There are several ways to divide data by train/validation or train/validation/test, but for the scope of experimentation in this thesis, I have only used the two-part division for simplicity. This task describe detail on the Fig 4.6

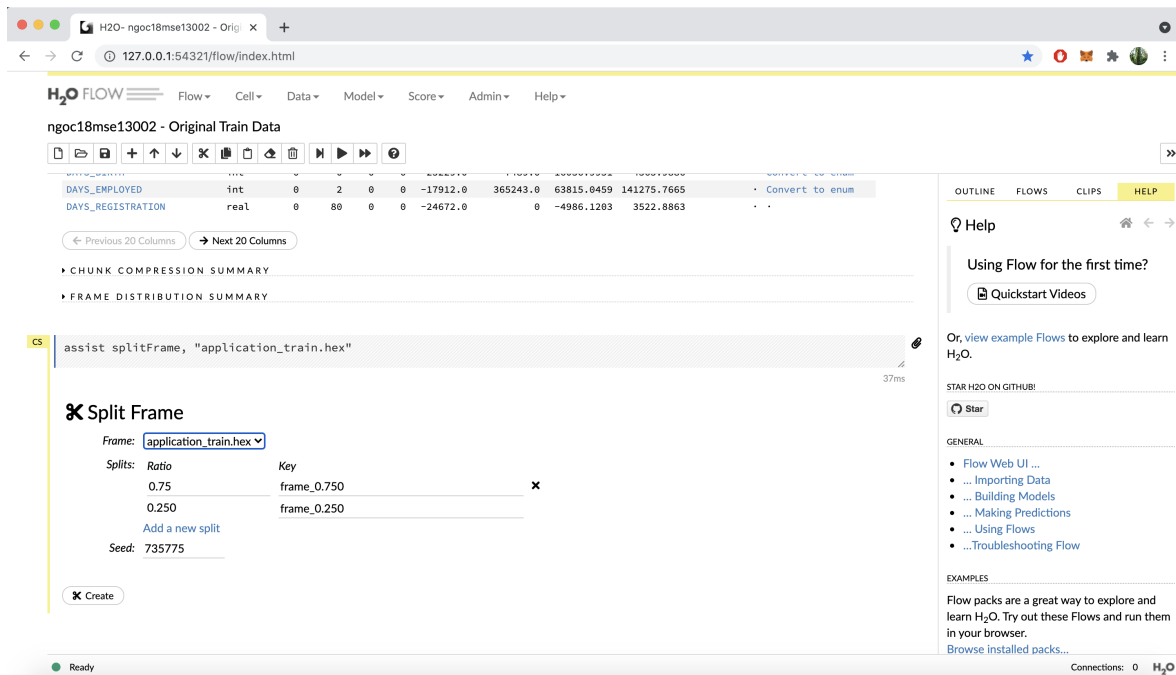


Figure 4.6: Split the train data

4.2.1.4 Step 4 : Train the AutoML model

After split the data to train set and test set, we move to next step Train Model. On this step, we can use H2O to build the model with specify algorithm or we can use AutoML. On the first experiment I will select AutoML because AutoML will automatically trains and tunes models with less parameters. I need to do only point out the train set and test set, identify the response column and specify a time constraint, a maximum number of models constraint, and early stopping parameters. On Fig 4.7 , we select *training_frame* and *valid_frame* split from section above, then select the *response_column* as *TARGET*. With other parameters such as *blending_frame*, *leaderboard_frame*, *project_name*, *fold_column*, *weights_column*, we can left it default value. On Fig 4.8 , we must select *sort_metric* as *AUC*. If we already selected the algorithm, we can remove the algorithms on the *exclude_algos* to save the training time. If we not decide to use which algorithms yet, we can left it default, H2O will run all algorithms or stop if meet the maximum number of models. Stacked Ensembles will also be automatically trained on the collection of individual models to produce a highly predictive ensemble model. With the remain parameter we can leave it as default value and then just click to Build Model button and waiting for training time. The default value for max run time of training individual model is 3600 second mean 1 hour. If we configure the parameter *max_runtime_secs_per_model*, the training time per each individual model can increase or decrease. On this first exeriment I was leave other parameters in default value, and training was completed after 1 hour.

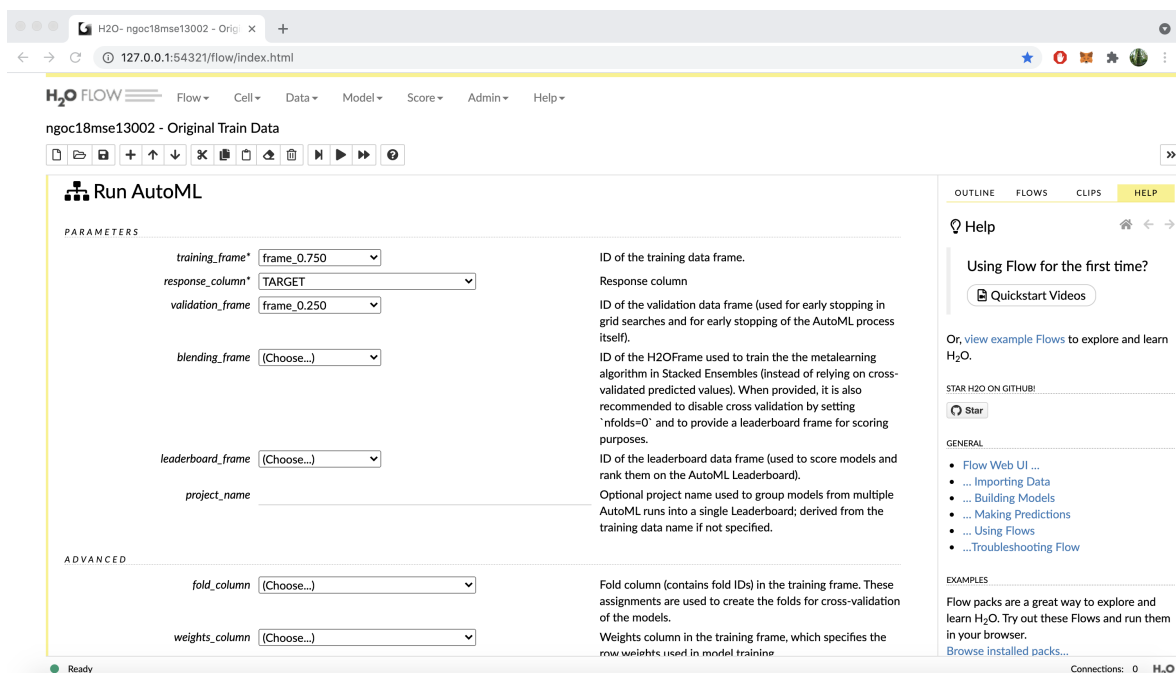


Figure 4.7: Train the AutoML model - Config Parameters 1

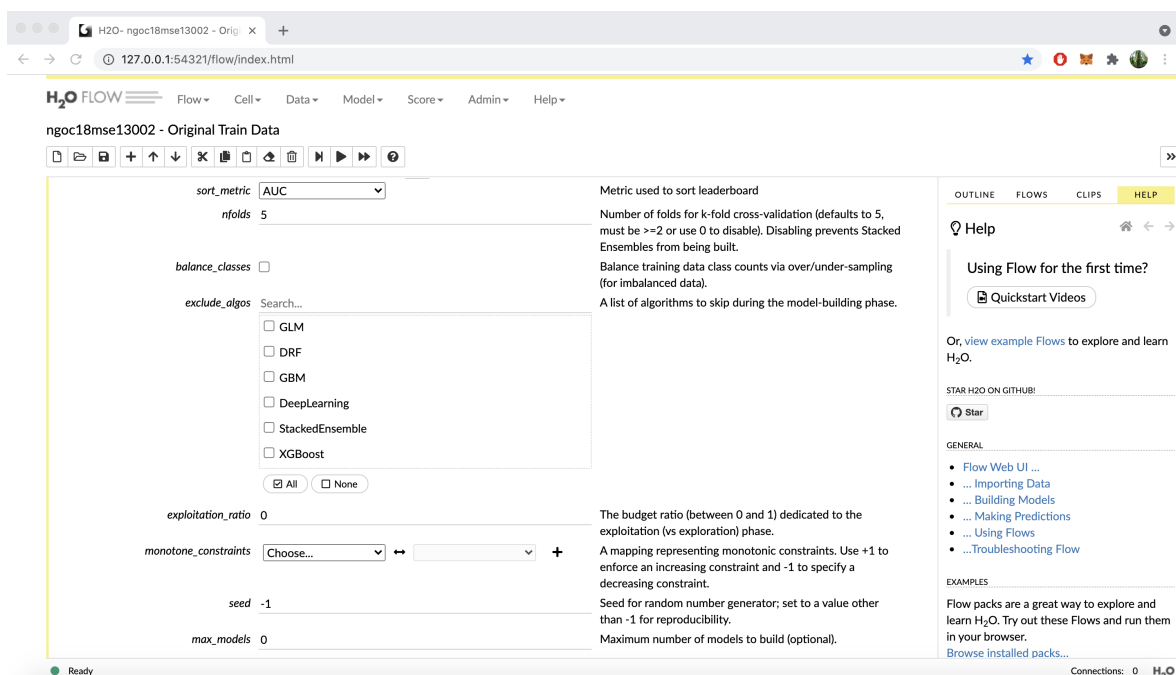


Figure 4.8: Train the AutoML model - Config Parameters 2

In case we choose to build the model using AutoML, H2O will train the Stacked Ensembles model based on other individual models. Stacked Ensembles model has high accuracy, in most cases is the highest in AutoML leaderboard evaluation of trained models. Model Stacked Ensembles is trained only for non-multiclass classification cases, for multiclass classification cases, only individual training models are generated.

Performance of models will show up on the leaderboard and the leaderboard will show the best results first. The data frame can be specified when configuring AutoML in the step above after splitting the data frame. If the leaderboard data frame is not specified, the default data frame will be the training frame[17].

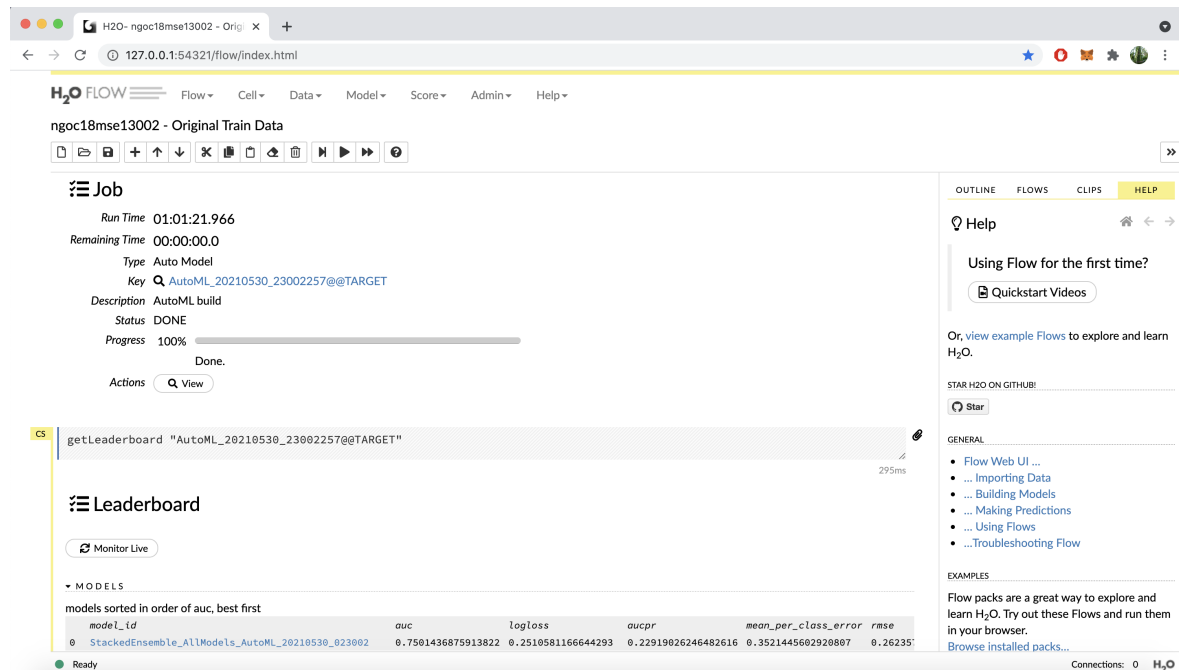


Figure 4.9: Train the AutoML model - Training completed

4.2.1.5 Step 5 : Review the leader board

After running the model, we can click the View button to see the leaderboard of AutoML H2O, the leaderboard is sorted according to the AUC configured in Fig 4.8 Config the Parameters 2. The models have the highest accuracy according to the AUC, routine are Stacked Ensembles models of H2O. Looking at the leader board, we can see that the 3 models with the highest accuracy are *StackedEnsemble_AllModels_AutoML_20210530_023002* with AUC 0.75 equivalent to GINI 0.5, second place is *StackedEnsemble_BestOfFamily_AutoML_20210530_023002* with AUC 0.7476 equivalent to 0.4952, and 3rd place is *XGBoost_3_AutoML_20210530_023002* with AUC 0.7476 equivalent to GINI 0.4918.

All that three models with the first test had relatively good accuracy. I have tried running some other algorithms like Random Forest, Gradient Boosting Machine on R Studio with this same training dataset but it only gives AUC from 0.6 to 0.7. Looking at the leaderboard, we can see that it is completely possible to use the XGBoost algorithm that will bring a relatively high accuracy, and if there is data processing, using all the data provided, it is possible to increase the accuracy. In addition, the points can be noticed that the algorithms using Grid Search also bring high accuracy, but I predict that due to the limitation of the maximum

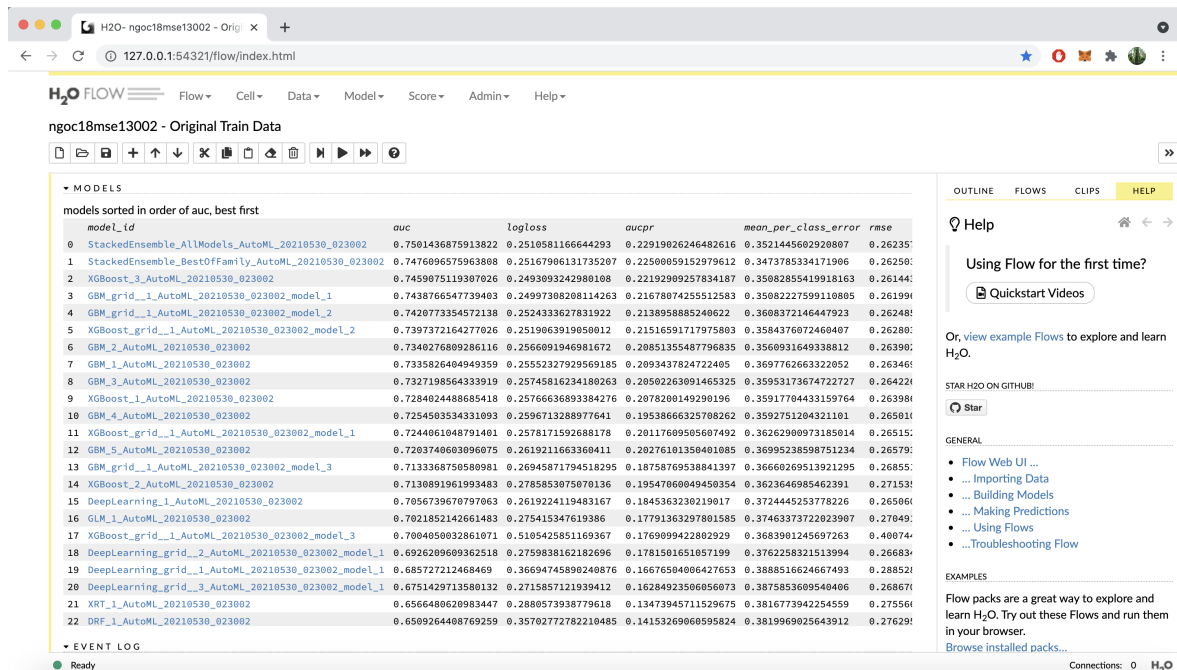


Figure 4.10: Review the leader board

time for independent models, the model training time is limited. short lead to the accuracy is not achieved as expected. Algorithms such as DRF - Distribution Random Forest, XRT - Extremely Randomized Trees, Deep Learning do not provide high accuracy with this training dataset. Thus, we have a first look at the training dataset that has been labeled from Home Credit when running first experimentally and without preprocessing as well as optimizing hyperparameters and features selection.

4.2.2 Second experiment : Performance with combined all data files

From the first experiment, we can easily see that choosing XGBoost algorithm for the specific problem Predict Default Risk from the dataset of Home Credit is completely appropriate. I still use H2O Flow instead of R Studio, import the data file that has combined all the files that Home Credit provides on Kaggle to build an XGBoost model. In this experiment, I will do not config the hyper parameters, just train the XGBoost model with default parameters. All steps for training will similar to the first experiment. In the first step, I will import data from a CSV file that was pre-processed and created on 4.1.2. In the second step, I will parse the train data to H2O data frame. In the third step, I will split the data frame into 2 data set with a ratio 0.75/0.25 for the training set and the validation set. Last step I will train the model. In this second experiment, because we have chosen the XGBoost algorithm suitable for the training data set and use default parameters, the training time will be short, about 10-12 minutes will be finished.

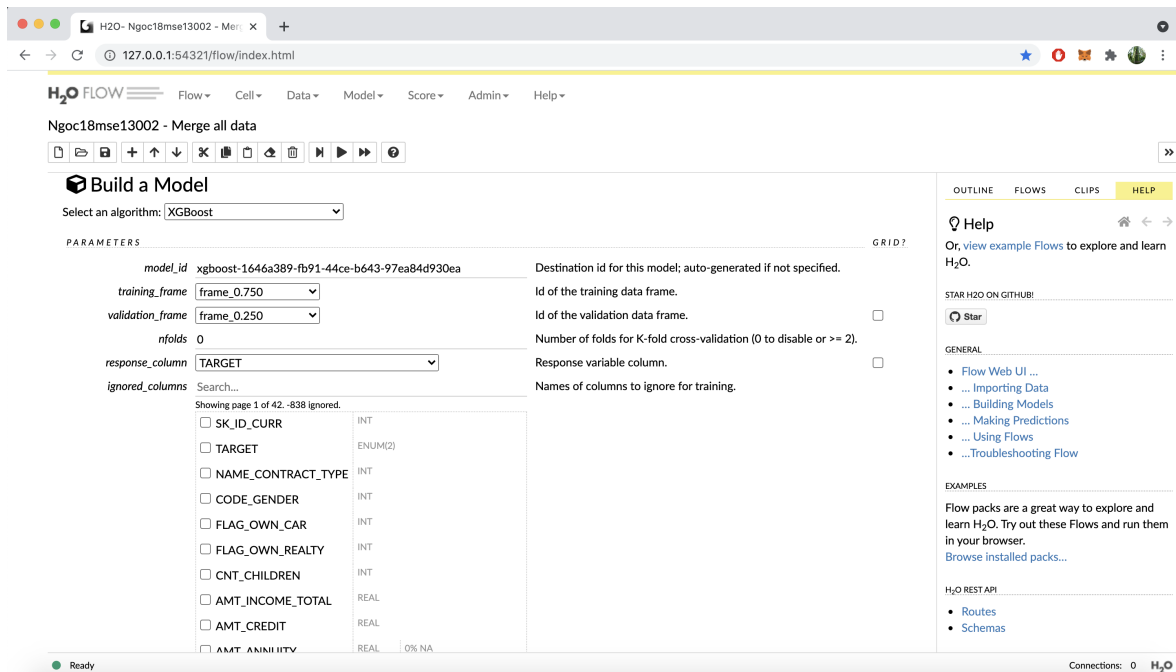


Figure 4.11: Train the XGBoost model

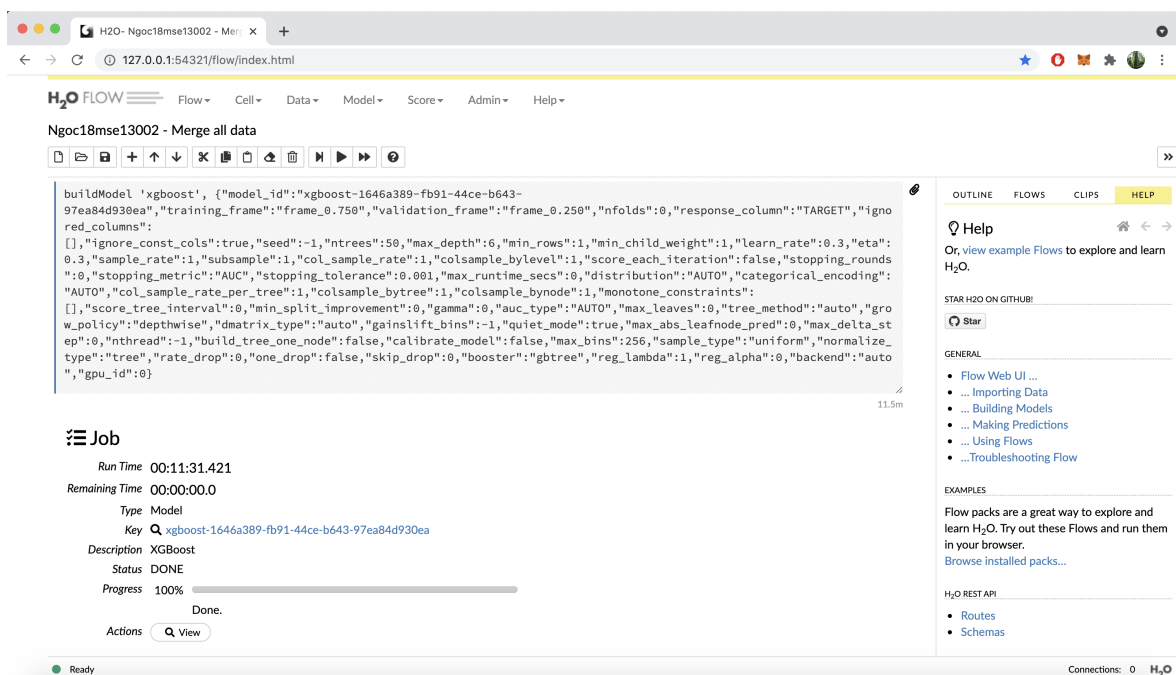


Figure 4.12: Train the XGBoost model - Training completed

After the training is finished, we can click the View button to see the details of the training process and the results achieved as well as some information such as Variable Importances sorted by image weight. affect the model. Since we built the model with the specified algorithm, there will be no leaderboard, and the model information will provide ROC Curves measured on the training and validation sets. Looking at Fig 4.13, we can see that the AUC

of the training set is 0.862 equivalent to GINI 0.724 and the AUC of the validation set is 0.77 equivalent to GINI 0.54. As such, the results have been significantly improved compared to the first test.

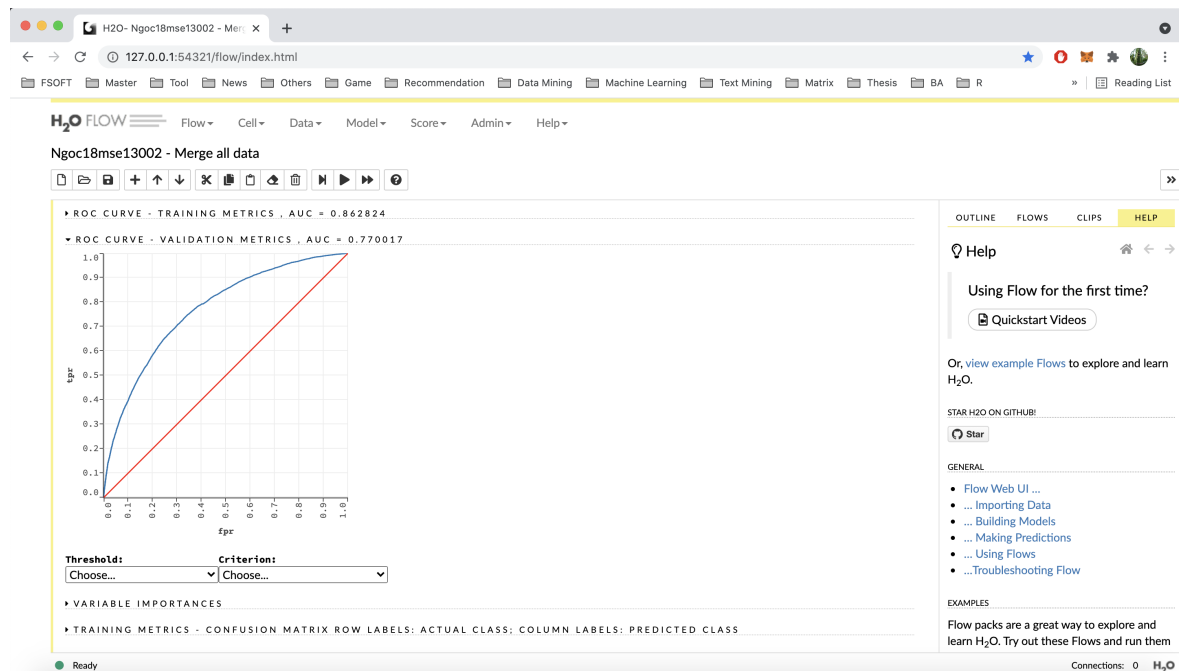


Figure 4.13: View the XGBoost model

4.3. Discussion

Thus, we have ended 2 times of experiments with clearly improved AUC results. In the second experiment, we have achieved the GINI is greater than or equal to 0.5 as mentioned in section 1.3 Research Purpose Statement. After achieving the goals of Section 1.3 for POC build time as well as achieving the require GINI index in the POC, we can review the information from the trained model to further improve the model for enhancing the performance and get high accuracy. Some information about the model can be seen in Fig Fig 4.14 as Variable Importances to know which are important features in the training data set, thereby removing the features that do not affect the model, helping to improve the accuracy and reduce training time. Looks into Fig 4.14 as you can see the most important features are *Ext_Source_3* and *Ext_Source_2* and *Ext_Source_1*. But the feature *Ext_Source_1* has a lot of missing objects in train data, we can consider trying to remove it to check the influence to model accuracy. This is the credit scoring of each customer from an external source. In Vietnam, we can know some other external source as CIC (Credit Information Center) [18] or credit scoring for personal customers from Transdata - FPT, etc...

▼ VARIABLE IMPORTANCES

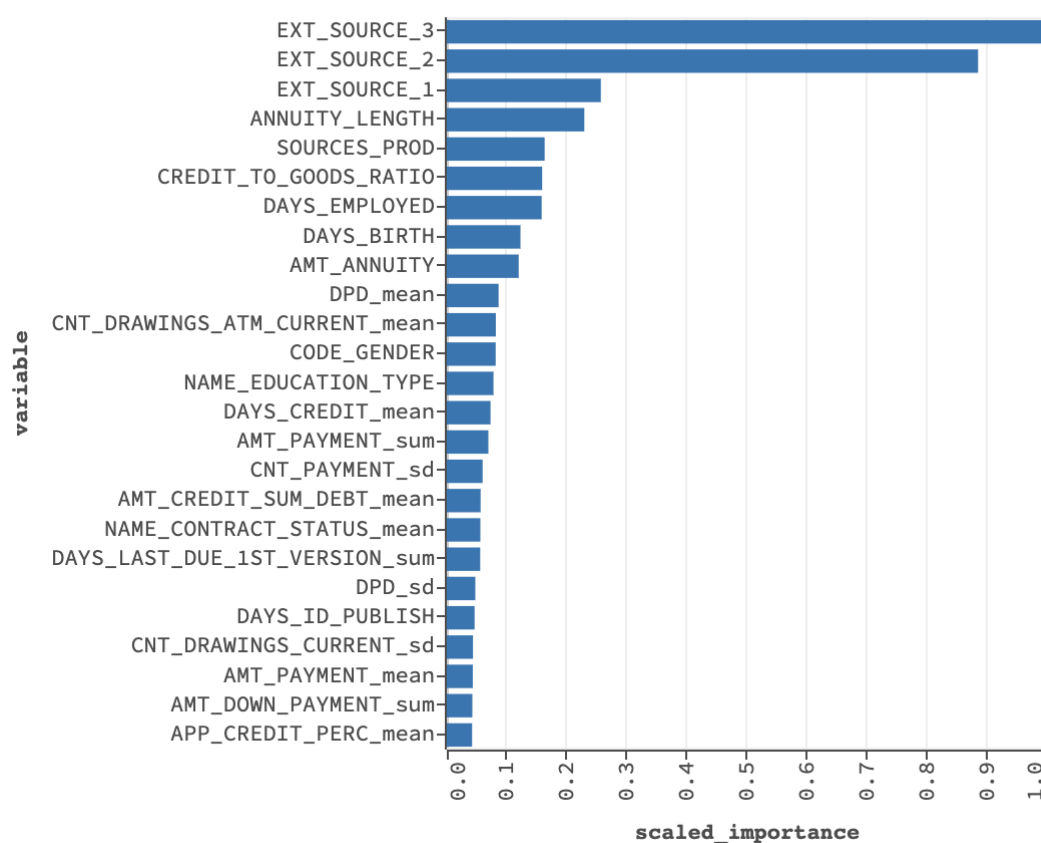


Figure 4.14: Variables Importances

H2O FLOW Flow ▾ Cell ▾ Data ▾ Model ▾ Score ▾ Admin ▾ Help ▾

Ngoc18mse13002 - Merge all data

response_column (Choose...)

ignored_columns Search...

Showing page 1 of 13. Filtered 125 of 419 - 419 ignored.

Column	Type	Missing
OWN_CAR_AGE	INT	66% NA
EXT_SOURCE_1	REAL	56% NA
APARTMENTS_AVG	REAL	51% NA
BASEMENTAREA_AVG	REAL	59% NA
YEARS_BUILT_AVG	REAL	67% NA
COMMONAREA_AVG	REAL	70% NA
ELEVATORS_AVG	REAL	53% NA
ENTRANCES_AVG	REAL	50% NA
FLOORSMIN_AVG	REAL	68% NA
LANDAREA_AVG	REAL	59% NA

☒ All ☐ None

Only show columns with more than 50 % missing values.

ignore_const_cols ☒

seed -1

ntrees 50

max_depth 6

Response variable column.

Names of columns to ignore for training.

Ignore constant columns.

Seed for pseudo random number generator (if applicable)

(same as n_estimators) Number of trees.

Maximum tree depth (0 for unlimited).

OUTLINE FLOWS CLIPS HELP

Help

Using Flow for the first time?

Quickstart Videos

Or, view example Flows to explore and learn H2O.

STAR H2O ON GITHUB!

Star

GENERAL

- Flow Web UI ...
- ... Importing Data
- ... Building Models
- ... Making Predictions
- ... Using Flows
- ... Troubleshooting Flow

EXAMPLES

Flow packs are a great way to explore and learn H2O. Try out these Flows and run them

Connections: 0 H2O

Figure 4.15: Unnecessary features

In the file `HomeCredit_columns_description.csv` describing the features of each data file Home Credit provides with the training data directory, we can easily see that there are many features that do not affect the customer's ability to pay the credit, such as flags indicating that the customer has submitted sufficient documents, or information about the building that the customer is living in such as how long it was built, how many elevators there are, etc... Looks in Fig 4.15, these characteristics are not important so the rate of missing values is very high, often over 50%. We can remove these features to reduce the model's training time and avoid causing noise to the model.

In addition, we can tune the hyper parameters to increase the accuracy of the model. In the next experiments, we can also use more data pre-processing techniques like OnHot Encoded or using Grid Search techniques to tuning the model. H2O Flow also has this option but needs to be tested on computers with strong configuration for building ML/AI models, if running on personal computers, it's not possible if the data set is too large like the Home Credit data set after combining all the data.

Chapter 5

CONCLUSION AND FUTURE WORK

In the scope of this thesis, I have used AutoML to predict the default risk in banking problems with Home Credit's data and have achieved the set goals of building a model quickly and simply. in a period of 2 weeks suitable for a PoC project and towards the goal of popularizing AI for employees in the banking and financial industries. AutoML is still a new field that is being continuously researched and developed, but in practice, it has brought effective results to meet the needs. The thesis has described in detail the process of building a default risk in the banking model in a simple, fast, and effective way. There are always differences with the research environment, so even though AutoML has its limitations, it can be applied in practice when the majority of the targeted audience are not experts in AI and science. compatibility of AutoML, specifically the H2O framework when integrating with cloud systems, big data used in banking systems. In the first experiment of the thesis, I have given a ranking of the accuracy of algorithms based on AutoML H2O from which it is possible to have an overview of the accuracy ratings of the algorithms for ease of use. easily choose which algorithm to use with the data set that Home Credit provides. In the second experiment of the thesis with the selection of the XGBoost algorithm and optimization of some hyper parameters and the removal of some features, it does not have a great impact on model building, avoiding model interference, consuming construction resources, the new model I built a model that has higher accuracy and is accepted in practice. So in the future, if we use support tools to explain the model, features like SHAP, Lime, etc., we can better understand the model and easily perform feature selection. Also, if provided with powerful mainframe computers for machine learning to run Grid Search [19] to tuning model or using the OneHot Encode[20] technical on processing data, I am fully confident that I can build more accurate models with the same data set. The default credit problem is a popular problem in the banking and finance industry and is still being researched and developed, using a lot of algorithms, there are many related studies. Applied AutoML H2O does not bring intensive effect but can bring initial effect for research and apply in practice, so I strongly believe that I can expand the scope of the thesis to make a research paper about Default Risk in banking. On another side, this research can develop and expand modules such as integrating into actual data mart systems, deploying in the cloud, providing interfaces and web services based on Flask and ShinyR to form a product that can be applied in practice at banks in Vietnam, that I am working with such as TPBank, VPBank, MSB, etc.

References

- [1] P. Agarwal, “Redefining Banking and Financial Industry through the application of Computational Intelligence,” *2019 Advances in Science and Engineering Technology International Conferences, ASET 2019*, pp. 1–5, 2019.
- [2] “History of artificial intelligence.” <https://qbi.uq.edu.au/files/40697/The-Brain-Intelligent-Machines-AI-timeline.jpg>, 2013. [Accessed: 2021-04-10].
- [3] M. Leo, S. Sharma, and K. Maddulety, “Machine learning in banking risk management: A literature review,” *Risks*, vol. 7, no. 1, 2019.
- [4] “Artificial intelligence vs machine learning.” https://aws.amazon.com/lex/?nc1=h_ls, 2019. [Accessed: 2021-04-10].
- [5] Dataiku, “Use Cases: Data Science, Machine Learning and AI in Banking,” 2020.
- [6] R. Duda, P. Hart, and D. G. Stork, *Pattern Classification*, vol. xx. 01 2001.
- [7] R. Polikar, “Ensemble learning,” *Scholarpedia*, vol. 4, no. 1, p. 2776, 2009. revision #186077.
- [8] L. Breiman, “Bagging predictors,” *Machine Learning*, vol. 24, no. 2, pp. 123–140, 1996.
- [9] T. Chen and C. Guestrin, “XGBoost: A scalable tree boosting system,” in *Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*, KDD ’16, (New York, NY, USA), pp. 785–794, ACM, 2016.
- [10] H. Jin, Q. Song, and X. Hu, “Auto-keras: An efficient neural architecture search system,” in *Proceedings of the 25th ACM SIGKDD International Conference on Knowledge Discovery & Data Mining*, pp. 1946–1956, ACM, 2019.
- [11] “Amazon lex - conversational ai for chatbots.” <https://towardsdatascience.com/artificial-intelligence-vs-machine-learning-vs-deep-learning-2210ba8cc4ac>, 2021. [Accessed: 2021-04-10].
- [12] J. H. Moore, “Tpot - automl.” <http://automl.info/tpot/>, 2021. [Accessed: 2021-04-10].

- [13] W. Fu, R. Olson, Nathan, G. Jena, PGijsbers, T. Augspurger, J. Romano, P. Saha, S. Shah, S. Raschka, sohn, DanKoretsky, kdarakos, Jaimecclin, bartdp1, G. Bradway, J. Ortiz, J. J. Smit, J.-H. Menke, M. Ficek, A. Varik, A. Chaves, J. Myatt, Ted, A. G. Badaracco, C. Kastner, C. Jerônimo, Hristo, M. Rocklin, and R. Carnevale, “Epistasislab/tpot: v0.11.5,” June 2020.
- [14] M. Feurer, A. Klein, K. Eggenberger, J. Springenberg, M. Blum, and F. Hutter, “Efficient and robust automated machine learning,” in *Advances in Neural Information Processing Systems 28* (C. Cortes, N. D. Lawrence, D. D. Lee, M. Sugiyama, and R. Garnett, eds.), pp. 2962–2970, Curran Associates, Inc., 2015.
- [15] T. Wei, C. Wang, and C. W. Chen, “Stable network morphism,” in *2019 International Joint Conference on Neural Networks (IJCNN)*, pp. 1–8, 2019.
- [16] E. LeDell and S. Poirier, “H2o automl: Scalable automatic machine learning,” *7th ICML Workshop on Automated Machine Learning (AutoML)*, July 2020.
- [17] H2O.ai, “Using flow - h2o’s web ui.” <https://docs.h2o.ai/h2o/latest-stable/h2o-docs/flow.html>, 2021. [Accessed: 2021-05-29].
- [18] T. S. B. of Vietnam, “National credit information center.” <https://cic.org.vn/>, 2021. [Accessed: 2021-05-30].
- [19] Google, “Overview of hyperparameter tuning.” <https://cloud.google.com/ai-platform/training/docs/hyperparameter-tuning-overview>, 2021. [Accessed: 2021-05-30].
- [20] L. Yu, R. Zhou, R. Chen, and K. K. Lai, “Missing data preprocessing in credit classification: One-hot encoding or imputation?,” *Emerging Markets Finance and Trade*, pp. 1–11, 10 2020.